



University of Frontier Technology, Bangladesh
Department of Internet of Things and Robotics Engineering

Course Code	CC 483
Course Title	Cloud Computing

CASE STUDY

Type 1 (Bare Metal) Hypervisor

Selection, Analysis & Cloud Deployment Strategy

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Case Study: How BrightCloud Solutions Chose a Type 1 Hypervisor

BrightCloud Solutions is a growing software company that provides online services to its customers. The company has around 150 employees and uses virtual machines to run its email system, customer database, and internal applications. At the beginning, the company used a Type 2 hypervisor installed on a Windows server because it was easy to set up and manage for their small infrastructure.

For the first few months, the system worked well. However, as the company expanded, more employees started using the applications and customer activities also increased. The IT team began noticing problems during busy hours. When many users accessed the system at the same time, the database became slower and some applications took longer to respond.

After checking the server, the IT team found that the main problem was the extra Windows operating system running below the hypervisor. The host operating system was already using CPU and memory resources, which reduced the resources available for virtual machines. This additional layer affected the overall performance of the virtual environment [1].

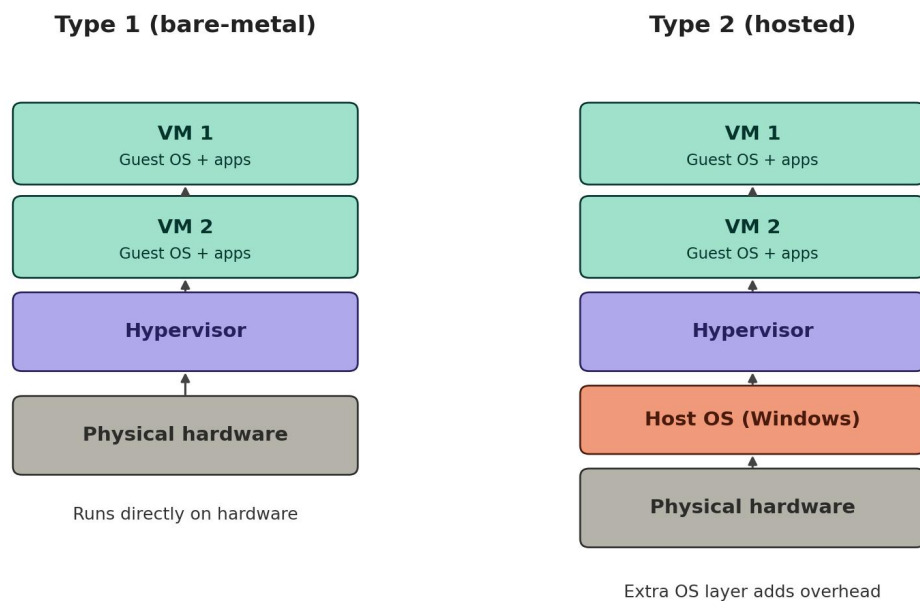


Figure 1: Type 1 (bare-metal) vs Type 2 (hosted) hypervisor architecture

The team also identified a security issue with the existing setup. Since all virtual machines were running on the same host operating system, any problem in the Windows system could potentially affect multiple virtual machines. The company needed a solution that could provide better separation between virtual machines and improve overall system reliability [2].

After reviewing different options, the IT manager, Mr. Karim, decided to replace the Type 2 hypervisor with a Type 1 hypervisor. The new hypervisor was installed directly on the physical server hardware instead of running on Windows. This removed the extra operating system layer and allowed the hypervisor to use server resources more efficiently [1].

After the migration, BrightCloud Solutions noticed several improvements. The database became faster during busy hours because virtual machines could use more available hardware resources. The company also achieved better isolation between virtual machines because they were no longer dependent on a shared host operating system.

The IT team was also able to manage the infrastructure more easily. They could run more virtual machines on the same servers and move virtual machines between servers during maintenance without affecting users. This helped the company maintain better service availability.

After six months, BrightCloud Solutions moved its infrastructure to a private cloud environment using Type 1 hypervisor technology. The company selected Type 1 instead of Type 2 because it provided better performance, stronger isolation, and more efficient use of hardware resources as the business continued to grow.

References

- [1] N. Chawla, A. Ghewari, S. Pawar, P. Thorat, R. Jarad, G. Dangare, and S. Pawar, “Hypervisor-Based Virtualization in Cloud Computing: Performance and Security Analysis,” *International Journal of Computational and Experimental Science and Engineering*, 2025, doi: 10.22399/ijcesen.2629.
- [2] S. Ally, “Assessment of Web Security Vulnerabilities for Common Open Source Virtualization Software,” *Tanzania Journal of Engineering and Technology*, vol. 43, no. 2, pp. 92–107, 2024, doi: 10.52339/tjet.v43i2.925.